CHE 396 Quiz 1

Huda Usman

TOTAL POINTS

79 / 100

QUESTION 1

1 1 **15 / 15**

- **- 3 pts** no detail explaination about the process
- **- 1 pts** no applications mentioned
- ✓ **- 0 pts** correct
- **- 15 pts** incorrect

QUESTION 2

2 2 **15 / 15**

- **- 12 pts** other processes not mentioned
- **- 3 pts** no brief information about the processes
- ✓ **- 0 pts** correct

QUESTION 3

3 3 **15 / 15**

- **- 10 pts** Major logical error in the BFD
- **- 5 pts** Minor mistake in the BFD
- ✓ **- 0 pts** correct

QUESTION 4

4 4 **15 / 15**

- **- 15 pts** seperation process not mentioned
- **- 10 pts** Partial process
- **- 5 pts** Minor error
- ✓ **- 0 pts** correct

QUESTION 5

5 5 **2.5 / 5**

- ✓ **- 2.5 pts** partial reaction
- **- 0 pts** correct
- **- 5 pts** incorrect

QUESTION 6

6 6 **2.5 / 5**

- **- 0 pts** Correct
- **- 5 pts** wrong technique
- ✓ **- 2.5 pts** partially correct

QUESTION 7

7 7 **3 / 5**

- **- 5 pts** No full form or production process
- **- 3.5 pts** full form but no production process
- **- 0 pts** correct
- ✓ **- 2 pts** partially production process
- **- 1.5 pts** no full form

QUESTION 8

8 8 **1 / 5**

- **- 0 pts** Correct
- **- 2 pts** Partially correct
- ✓ **- 4 pts** wrong
- **- 5 pts** not written

QUESTION 9

9 9 **10 / 20**

- **- 10 pts** incorrect logic for recovery rate (step 1)
- **- 10 pts** incorrect logic for recovery rate (step 2)
- **- 0 pts** correct
- ✓ **- 5 pts** correct logic but calculation error for recovery rate (step 1)
- ✓ **- 5 pts** correct logic but calculation error for recovery rate (step 2)

CHE 396 SENIOR DESIGN Quiz 09/24/2019

Huda Usman

Name

1. What is the design basis for your mini-project?

↳ assuming 350 production days

↳ $\frac{45{,}000\ O_2\ \text{ton}}{\text{day}} \Big| \frac{350\ \text{production days}}{1\ \text{year}} = 15750000\ \frac{\text{ton}\ O_2}{\text{year}}$

↳ design basis: 10% of 15750000 ton O_2/year = $1575000\ \frac{\text{ton}\ O_2}{\text{year}}$ will be produced.

$1575000\ \frac{\cancel{\text{ton}}\ O_2}{\text{yr}} \times \frac{2000\ \text{lb}}{1\ \text{ton}} \Big| \frac{1\ \text{yr}}{350\ \text{days}} \Big| \frac{1\ \text{day}}{24\ \text{h}} \Big| \frac{1\ \text{hr}}{3600\ \text{s}} = 104.16\ \text{lb/s}\ O_2$ produced

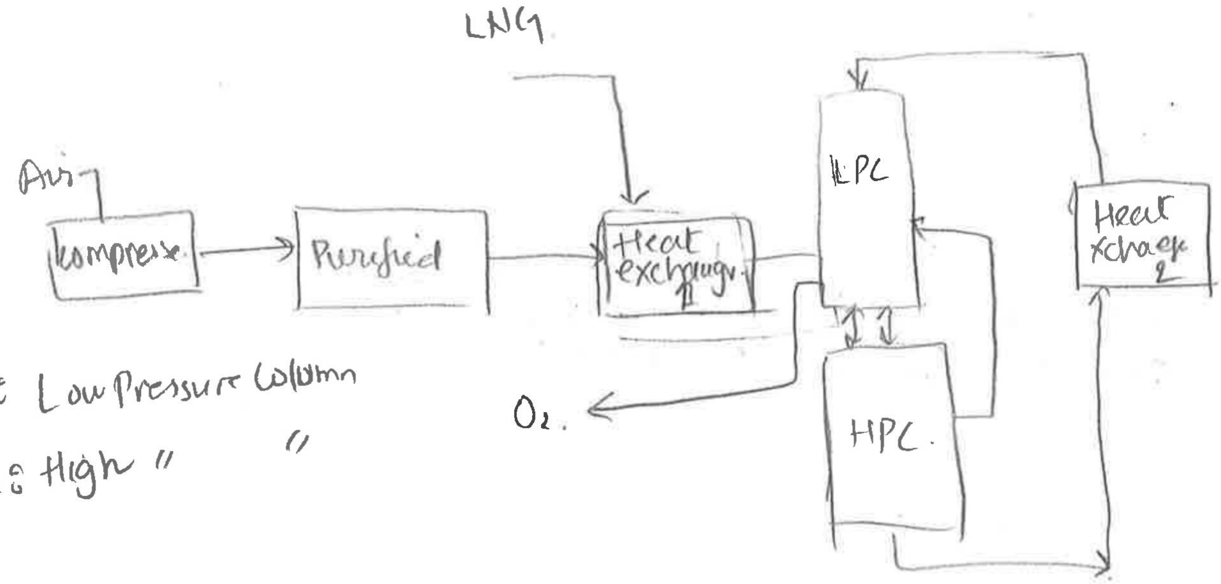
↳ $0.9999(x) = 1575000 \Rightarrow x = 1575157$ ton air will be supplied.

$1575157\ \frac{\text{ton air}}{\text{year}} \times \frac{1\ \text{yr}}{350\ \text{days}} \times \frac{1\ \text{day}}{24\ \text{h}} \times \frac{1\ \text{h}}{3600\ \text{s}} \times \frac{2000\ \text{lb}}{1\ \text{ton}}$ year $= 104.17$ lb/s air must be supplied to produced O_2 with 99.99% conversion

2. List competing processes.

1. Adsorption swing Process.
2. Adsorption through vacuum.
3. membrane separation
4. Separation through flash chamber.
5. conventional cryogenic separation

3. Draw BFD for your mini-project



LPC: Low Pressure Column

HPC: High " "

4. Explain separation section of your mini-project in details.

Pre-cooled air enters the High pressure coloumn. where air is seperated into its components N_2 and O_2.

O_2's Boiling point is $-183°C$ and in HPC it settles down and N_2 gas evaporates because of lower boiling of about $-192°C$ and is collected from the top of the coloumn.

The oxygen-rich liquid product of HPC is then passed all the heat exchanger #2 and is fed to the lower pressure column as feed to further purify the O_2 from impurities due to the differences in boiling points.

At the end, the pure O_2 is obtained from the bottom of the LPC and is then compressed to increase the flow.

Runing pressures. HPC ~ 130kPa., # of stages 50

LPC ~ 0 KPa, # of stages 60.

of stages.

5. Write down the chemistry of H_2SO_4 production process.

$SO_3 + H_2O \longrightarrow H_2SO_4$

6. Explain how paraxylene is separated from other xylenes (ortho and meta)?

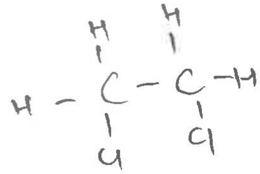
In distillation colomns ~~by~~ increasing the # of stages since they both have similar cphysical properties

7. What is LNG? Explain briefly its production.

LNG is Liquified Natural gas. It's main component is methane. It scan be found naturally but then to generated useable LNG energy - it goes through many separating techniques

8. How oxygen is produced in chemical industry?

Cryogenic air separations techniques mostly.



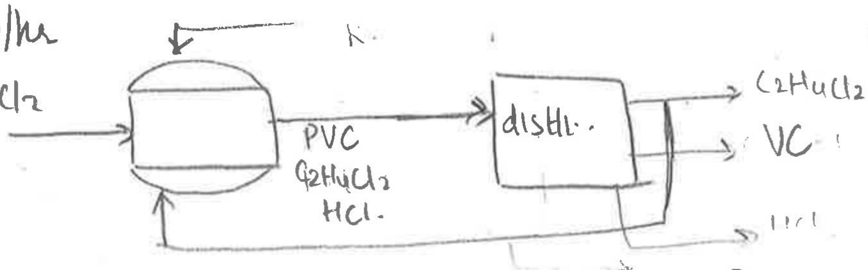
$C_2H_4Cl_2$ $C_2H_3Cl_3$

9. Dichloroethane (90,000lb/hr) enters pyrolysis unit. Reaction happens with a conversion of 70% producing vinyl chloride and HCl. After reaction dichloroethane is separated by distillation with a recovery rate of 95% and recycled back to the reactor. Calculate the vinyl chloride output flow rate and recycle stream flow rate in lb/hr.

conversion = 0.70

$C_2H_4Cl_2 \rightarrow C_2H_3Cl + HCl$

90,000lb/hr
$C_2H_4Cl_2$



Recovery (0.95)

$0.7(90{,}000\ \frac{lb}{hr}) = 63{,}000\ \frac{lb\ VC}{hr} \Rightarrow$

$0.95(27{,}000\ \frac{lb}{hr}) = 25650\ \frac{lb\ C_2H_4Cl_2}{hr}$ is recovered

$(1-0.7)(90{,}000\ \frac{lb}{hr}) = 27{,}000\ \frac{lb\ VC}{hr}$

$0.7 = \frac{90{,}000\ lb/hr - out}{90{,}000}$

$63{,}000 = 90{,}000 - C_2H_4Cl_2(OUT)$

$\boxed{C_2H_4Cl_2(OUT) = 27000\ \frac{lb}{hr}\ C_2H_4Cl_2}$ out of the reactor.

dist	
VC	$C_2H_4Cl_2$ = 99 g/mol lb/lbmol
	C_2H_3Cl = 62.5
	HCl = 36.5

Outflow;

VC: $\frac{25650\ lb\ C_2H_4Cl_2}{hr\ 99\ lb/lbmol} \times 62.5\ \frac{lb\ VC}{lbmol} = 16193\ \frac{lb\ VC}{hr}$

R: $(X+R) \times 0.95 = R$

$25650(0.95) + 0.95R = R$

$24367.5 + 0.95R = R \Rightarrow 24367.5 = 0.05R$

$\boxed{R = 487350\ \frac{lb}{hr}\ \text{recycle}}$